

Energy-efficient Analog Integrated Circuits

DCDC Converters – Assignment 3

Assignment

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Introduction

In this assignment you will learn how to design a Successive-Approximation Switched-Capacitor DCDC converter topology (SAR), an attractive solution for several IoT applications where miniaturization and fine-grained conversion ratio resolution are required.

Please, refer to the following paper, as well as the course lecture notes:

S. Bang, D. Blaauw and D. Sylvester, "A Successive-Approximation Switched-Capacitor DC–DC Converter With Resolution of $V_{\text{IN}}/2^N$ for a Wide Range of Input and Output Voltages," in IEEE Journal of Solid-State Circuits, vol. 51, no. 2, pp. 543–556, Feb. 2016, doi: 10.1109/JSSC.2015.2501985.

The exercise will focus on the power stage and the gate drive schemes, while neglecting the phase generation and the control scheme for simplicity. You will not have to design the converter exactly as presented in the paper, please, follow the specifications listed below to choose the appropriate number of stages, stage topology, switch scheme, and so on.

Specifications of the Successive-Approximation Switched-Capacitor Step-Down DCDC Converter

ILOAD=0

VIN	2.4V
VOUT_nominal	0.15V:0.15V:2.4V
VOUT_resolution_max	150mV
C_tot	5nF

ILOAD=100uA

VOUT_nominal	EFF_peak	ΔV_{OUT}
0.6V	>75%	<15mV
1.2V	>80%	<15mV
1.8V	>85%	<15mV

INSTRUCTIONS AND SUGGESTIONS

Switched-Capacitor Power Stage

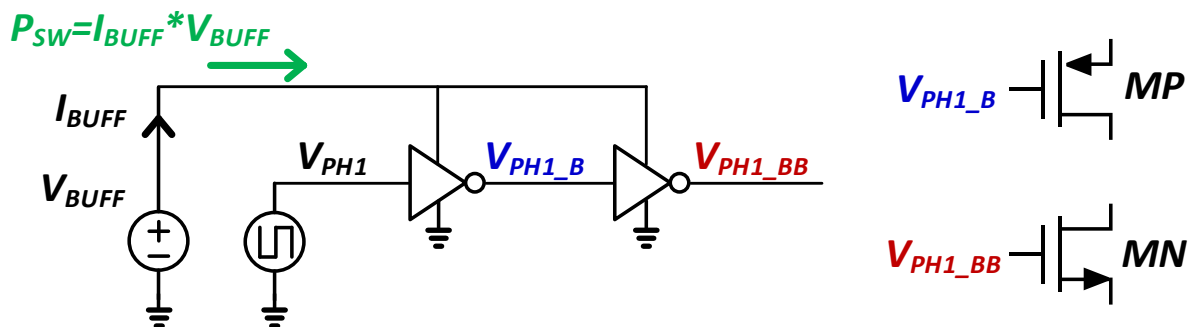
The SAR DC-DC converter is composed of several cascaded stages with configurable VH and VL at each stage. In the paper a total of 6 stages are cascaded, the first one being a 4:1 ladder and the other 5 being 2:1 interleaved 2:1 SC. You are not required to build it following such structure, number and topology of the stages. Please, choose the number of stages based on the VOUT range and resolution specifications and feel free to cascade several identical 2:1 interleaved SC stages.

The given $C_{\text{tot}}=5\text{nF}$ is the maximum value of capacitance you can use to implement all capacitors in your circuit. Thus, not only the flying capacitors in each stage but also the eventual output decoupling capacitor as well as the level shifter capacitors. It is recommended to use the `mimcap_3t` from the `tsmcN65` library. If you choose a different capacitor type you will have to clearly motivate why and to make sure that the model includes the bottom-plate parasitic capacitance, which is typically the case in 3 terminals (*_3t) components.

Pay attention to each device voltage rating. You can use core devices with a maximum rating of 1.2V (`nch` and `pch`) or IO devices with a 2.5V rating (`nch_25` and `pch_25`). Voltages higher than those rating applied to the Vgs or Vds will be dangerous for the circuit operation and will be synonyms of a bad design. The TSMC 65nm process allows deep-nwell to isolate a p-type bulk of an NMOS and connect it to a different voltage than GND. Thus, feel free to tight the bulk to the source of all NMOS transistor, as well as PMOS ones.

Phase Generation

Please, use ideal vpulse generators from the `analogLib` to create your 2 non-overlapped phases similarly to previous exercises. To easily estimate the converter switching losses, it is recommended as a good practice to only use buffered signals inside your blocks. This means that the voltage generated by the vpulse will only connect to a set of inverters which power will be measured to estimate the switching losses, as in the figure below. You do not need to use any library inverter, please, make your own so that you can play with the sizing for signals that need a certain driving strength.



Level Shifting

The two vpulse generators for the phases A and B are the only ideal sources allowed to drive the transistor gates. As explained in the paper as well as the lecture notes, NMOS and PMOS capacitive level shifters can be designed to drive the power transistor gates.

Voltage Rails and ideal components allowed

Another ideal source vdc from the analogLib will be used to generate the input voltage VIN.

For the eventual logic rails at the IO voltage of 2.5V you can directly use the input voltage to include their power into consideration.

For eventual logic rails at the core voltage of 1.2V you are allowed to use one additional ideal vdc, as long as you include its power into consideration in your efficiency calculation.

Furthermore, you can generate the control voltage of the configuration switches to select the SAR ratio with ideal voltage sources.

Finally, in your top testbench you will have to model the load current with a R_load resistor or with a ILOAD idc current generator, as in previous exercises. However, for the load capacitor use the mimcap_3t and make sure it is included in the maximum allowed 5nF.

Feel free to use more ideal sources in an early design phase of your circuit to better understand the circuit behaviour or to distinguish between different loss mechanisms.

However, in the final version to be delivered you should only have the ideal components listed above.

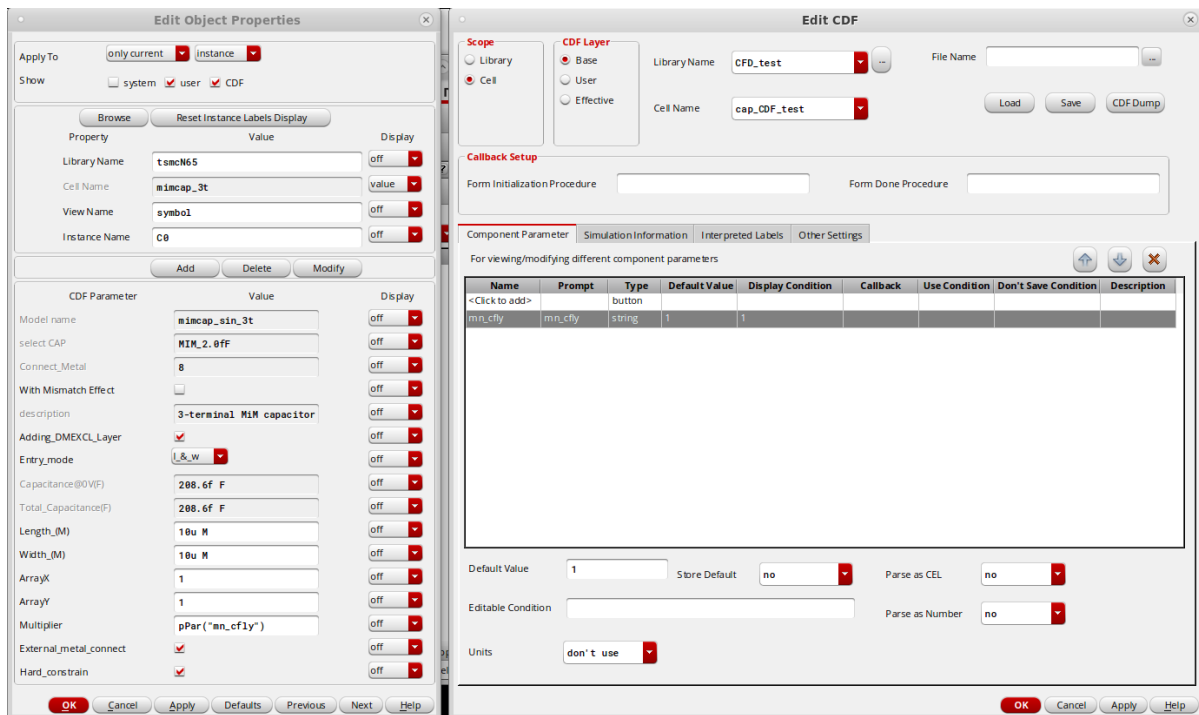
The same goes for other components, if you see some unclear behaviour you can investigate it simulating with ideal flying capacitors, then add an ideal bottom-plate capacitance and later on move to the real mimcap_3t, as an example.

CDF parameters

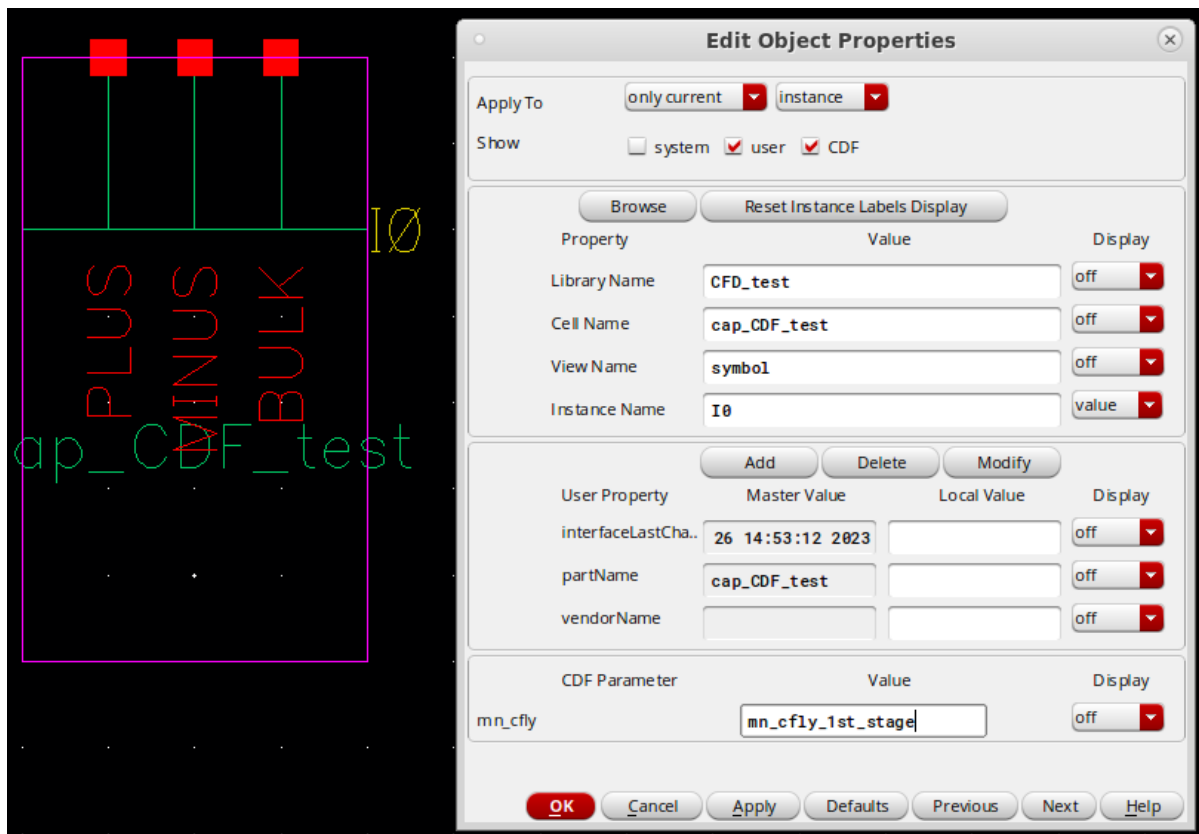
The CDF parameters can be used when you have instantiated multiple identical cells but you want to size one parameter of each cell differently from each other. For instance, if you instantiate N identical SC cells but you want a different value of CFLY.

Let's say that you want to parametrize the multiplier of your mimcap_3t flying capacitor:

- Open its properties with “q” and write pPar(“mn_cfly”) in the Multiplier entry, then Check+Save the schematic.
- From the CIW: Tools → CDF → Edit
- Then enable Base as CDF layer, find the interested cell and introduce the “mn_cfly” as follows. Use string as a type so that you will be able to parametrize with a global variable later. Also, put a default value.



Then, create a symbol of the parametrized unit and when you instantiate it you will be able to assign a variable to that specific instantiation. Thus, you can instantiate N identical symbols and assign a different variable to each one of them.



The use of pPar and CDF parameters in this way works only with 1 level of hierarchy. Therefore, make sure to introduce the component you want to parametrize inside the cell where you want to set the CDF parameter.

Transient and PSS simulations

The transient simulation is an obvious choice to check the behaviour of the interested signals over time. However, with an increasing number of components, high frequency switching signals and a long time to reach steady-state, it might not be the fastest way.

When you want to analyze a circuit with periodic operation in steady-state, a Periodic Steady-State (PSS) simulation might be handy. For a preliminary check of the circuit behaviour, the simulation can be run with liberal accuracy without specifying a maxstep. Then, you might want to increase the accuracy to compute current components when calculating the power consumption.

However, the writing of the expression is a little bit more tricky than a simple transient simulation. If you want to observe a voltage over the time period as a result of the pss simulation, you can type the following:

- `vtime('pss "/VOUT")`

For a current it would be:

- `itime('pss "/V0/MINUS")`

You can write those expression in your output setup, name them and use the name to compute their `average()`, `abs()`, `ymin()`, `ymin()` or any other required operation.

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|---|-------------------------------|--|----------------------------------|
| ▪ | <code>VOUT_pss</code> | | <code>vtime('pss "/VOUT")</code> |
| ▪ | <code>VOUT_average_pss</code> | | <code>average(VOUT_pss)</code> |

QUESTIONS

- 1. In your report, explain the following features of your design on Cadence:**
 - a. Choice of the Number of cascaded stages
 - b. Topology of each stage
 - c. Gate driving scheme from the ideal vpulse to the power transistor gates
 - d. Choice of the transistor types and rating for the power transistors, configuration switches and level shifters
 - e. Allocation of the total 5nF capacitance
 - f. Sizing of the power transistors, configuration switches and level shifters
 - g. How the efficiency has been computed
 - h. Detail which specifications your design manages to accomplish and whether some are not fulfilled. In such case, please, explain how you proceeded.
- 2. Provide the following plots and data from your simulation results on Cadence:**
 - a. Transient waveforms of VIN, VOUT, each intermediate VH, VM and VL node, and both generated phases, when providing VOUT=1.2V with ILOAD=0A
 - b. A table printed from the cadence calculator of the average VOUT sweeping all the configurations when ILOAD=0A to show all the possible VOUT values and the desired VOUT maximum granularity of 150mV.
 - c. An efficiency plot of the three configurations providing a nominal VOUT of 0.6V, 1.2V and 1.8V, respectively, with ILOAD from 0 to 100uA, showing the efficiency profile and its maximum value in all three configurations. Please, add a printed table as well with the efficiency and the VOUT_ripple specifically at ILOAD=100uA.
- 3. Provide the path of your final testbench folder:**
 - a. Make sure that the provided testbench is plug and play, set on the configuration of point 2c, thus, providing the efficiency plots just by running and plotting the results.
 - b. Please, explain how you planned your testbench and what settings need to be changed to check the results as in the previous points 2a, 2b and 2c.